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Trends in Distribution of Plains Minnow (Hybognathus placitus) in Kansas from 1964 to 2017

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ABSTRACT.—Streams of North America have been degraded by over a century of anthropogenic stressors, including channelization, fragmentation, impoundment, dewatering, and the modification of the natural flow regime. These stressors have contributed to the decline of many freshwater fish species and are particularly detrimental to the pelagic-broadcast spawning reproductive guild of cyprinids. Members of this guild require an unfragmented river section of 458 (±137) river kilometers (rkm) in length to persist. The Plains Minnow is a member of this reproductive guild and is listed within Kansas as a threatened species. Plains Minnow were once common throughout Kansas, but have experienced declines in abundance and distribution. Using historical data and 14 y of new data, we analyzed the proportion of sites occupied by Plains Minnow over time within Kansas as a whole and within specific river basins using parametric linear regression models and nonparametric localized regression models. We found Plains Minnow to be declining within the historical range of the species across the state as a whole and specifically within the Smoky Hill-Saline and Lower Arkansas river basins. We found Plains Minnow distributions to be unchanging within the Kansas-Lower Republican and the Cimarron river basins. Within the Smoky Hill-Saline and the Kansas-Lower Republican river basins, the proportion of sites occupied by Plains Minnow has been at or near zero for the majority of the time period studied. Our findings support previously published studies and contribute to the growing body of evidence that Plains Minnow continue to decline within Kansas. If the species is to recover within the state, anthropogenic stressors will have to be mitigated, and repatriation efforts will need to be conducted.

INTRODUCTION

Streams of North America have been subjected to over a century of anthropogenic stressors leading to the degradation of river systems and the imperilment of approximately 21.3% of fish species native to the continent (Cross *et al.*, 1985; Williams *et al.,* 1989; Samson and Knopf, 1994; Nislow *et al.*, 2011; Perkin and Gido, 2011). Anthropogenic stressors negatively impacting North American fishes consist of channelization, fragmentation, impoundment, dewatering, and the modification/regulation of seasonal flows (Benke, 1993; Dynesius and Nilsson, 1994; Perkin *et al.,* 2015a). Ricciardi and Rasmussen (1999) determined that within the 100 y prior to their study, the natural rate of extinction for freshwater fishes was 0.4%, and inferred that over the next 100 y, the extinction rate of freshwater fishes of the North American continent will be near 2.4% or greater. More recent estimates of extinction rates for riverine fishes range from 312–2052% higher than the rate of natural extinction (Burkhead, 2012; Tedesco *et al.*, 2013).

Streams of the Great Plains have experienced many of the aforementioned stressors and have been degraded as a result. Benke (1990), using the Nationwide Rivers Inventory Database, analyzed 1524 rivers within the United States and determined only 42 of these rivers contained segments greater than 200 km free of fragmentation. Within the Great Plains, over 19,000 dams disrupt the connectivity of rivers and regulate the flow of water (Cooper, 2013). Compounding the reduction of connectivity created by dams, water extraction, drought, and reservoir operations have led to the dewatering and desiccation of

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streams within the region (Falke *et al.*, 2011), which alters the fish communities present within these stream reaches (Perkin *et al.*, 2015a; Perkin *et al.*, 2017). Additionally, in some areas of the Great Plains, water is being lost due to the exceptionally high evapotranspiration rates of Saltcedar (*Tamarix spp.),* an invasive plant species becoming more common in the region (Di Tomaso, 1998). At high densities, Saltcedar can lower the water table within a locality leading to dewatering of the stream channel (Di Tomaso, 1998). Furthermore, over 90% of land area within the Great Plains has been transformed from native prairies to agriculture (Dodds *et al.*, 2004; Gido *et al.*, 2010). The combination of the aforementioned stressors is detrimental to endemic Great Plains fish assemblages, as these assemblages have evolved to survive in wide sandy rivers of variable flow and not the fragmented and heavily modified rivers that are characteristic of the region today (Cross *et al.*, 1985; Perkin *et al.*, 2015a; Pennock, 2017). Hoagstrom *et al.* (2011) determined 84% of the 49 endemic Great Plains fish species have experienced declines or extinction due to anthropogenic modification of the aquatic environments of the Great Plains region.

The pelagic-broadcast spawning reproductive guild of Great Plains cyprinids has experienced marked declines in abundance and distribution as a consequence of the anthropogenic modifications of river systems in the region (Perkin and Gido, 2011; Perkin *et al.*, 2015b; Pennock, 2017; Worthington *et al.*, 2018). Species of the reproductive guild found within Kansas include the Plains Minnow (*Hybognathus placitus),* Peppered Chub (*Macrhybopsis tetranema*), Arkansas River Shiner (*Notropis girardi*), Sturgeon Chub (*Macrhybopsis gelida*), Shoal Chub (*Macrhybopsis hyostoma*), Sicklefin Chub (*Macrhybopsis meeki*), and Silver Chub (*Macrhybopsis storeriana*; suspected pelagic-broadcast spawner) (Gido *et al.,* 2010; Perkin and Gido, 2011; Kansas Fishes Committee, 2014; Perkin *et al.*, 2015b). The characteristic reproductive method of this guild involves the broadcast into the water column of semi-buoyant eggs that proceed to develop as they are carried downstream (Moore, 1944; Cross, 1967; Miller and Robison, 1973; Platania and Altenbach, 1998; Perkin and Gido, 2011). Eggs hatch after 24–28 h and continue to drift for 2–3 d as larvae develop (Lehtinen and Layzer, 1988; Platania and Altenbach, 1998; Perkin and Gido, 2011). Pelagic-broadcast spawning cyprinids are particularly vulnerable to dewatering and fragmentation (Cross *et al.*, 1985; Dudley and Platania, 2007). Perkin and Gido (2011) determined that for populations of pelagic-broadcast spawning cyprinids to persist, an unfragmented river section of 458 ($\pm$137) rkm in length is necessary depending on the species.

The Plains Minnow is a short-lived cyprinid (most individuals die during their second summer; Taylor and Miller, 1990), with adults growing to about 130 mm total length (Taylor and Eberle, 2014). The species can be found throughout the Great Plains from southern Saskatchewan and North Dakota to Texas (Sylvester *et al.,* 2005). Plains Minnow are adapted to live in perennial streams characterized by shallow waters and braided channels, and are typically found in areas of lower current velocity, such as backwaters and pools around sandbars (Cross, 1967; Platania and Altenbach, 1998). Worthington *et al.* (2018) noted that Plains Minnow were found in association with water velocities of 0.05 m s^{-1}. Plains Minnow spawn periodically throughout the summer, with spawning events generally synchronized by high flow events (Sliger, 1967; Lehtinen and Layzer, 1988; Durham and Wilde, 2006). During the spawning season, Plains Minnow aggregate to spawn in areas of lower current velocity, such as backwaters and alongside sandbars (Cross, 1967; Taylor and Miller, 1990). While Plains Minnows are known to choose habitat types with lower current velocity during spawning, it is vital that there is some water movement present within the spawning habitat and downstream river segment as their semi-buoyant eggs must remain suspended in the water column as they develop (Coleman, 2015; Worthington *et al.,* 2018).

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TABLE 1.—Past studies analyzing the status of Plains Minnow within the state of Kansas either at the state level or within specific river basins. Studies in which dates of sampling or year range for data were not given are indicated with an asterisk (*)

Basin	Date (of sampling)	Status	Author
State of Kansas	1885–1995	Declining	Eberle *et al.* (1997)
	2005*	Declining	Haslouer *et al.* (2005)
	2007*	At Risk of Extirpation	Eberle *et al.* (2007)
Arkansas	1947–2003	Declining	Gido *et al.* (2010)
Lower Arkansas	1950–2013	Declining	Perkin *et al.* (2014)
Upper Arkansas	1993*	Likely Extirpated	Eberle *et al.* (1993)
Kansas	1947–2003	Declining	Gido *et al.* (2010)
	1993*	Rare	Wenke *et al.* (1993)
Smoky Hill-Saline	1947–2003	Declining	Gido *et al.* (2010)
	1966*	Rare	Summerfelt (1967)

Throughout Kansas, Plains Minnow were once abundant in all large streams with wide channels of sand and shallow braided flow (Cross, 1967; Cross and Collins, 1995). However, due to anthropogenic stressors, the species has experienced declines in abundance and distribution throughout the state (Haslouer *et al.*, 2005; Table 1). Five years ago, the species was considered rare within the state, except within the Cimarron, Medicine Lodge, and Salt Fork Arkansas river basins (Taylor and Eberle, 2014). Haslouer *et al.,* (2005) proposed that it would be more appropriate for Plains Minnow to be listed as an endangered species within the state due to range constrictions, extirpations, and major habitat changes within their range. However, 14 y after Haslouer *et al.*, (2005) proposed listing the species within Kansas as endangered, the status of Plains Minnow is still listed as threatened by the Kansas Department of Wildlife, Parks, and Tourism (KDWPT) (Kansas Fishes Committee, 2014).

Our study builds upon previous studies (years 1973–2005 Haslouer *et al.,* 2005; years 1947–2003 Gido *et al.*, 2010) by looking at historical trends and including 14 new years of data through 2017. While multiple studies highlight the decline of Plains Minnow within various basins of the state, there are no studies that analyze the distribution of the Plains Minnow within the entire historical range of the species in Kansas. The purpose of this study was to analyze trends in distribution of Plains Minnow across the historical range of the species in Kansas and within specific river basins in the state using historical as well as new data collected by the KDWPT Stream Survey 2004–2017.

METHODS

DATA COLLECTION

Data used in our study was collected from the KDWPT Ecological Services Section Stream Survey Database and supplemented by a review of pertinent literature. The database is a compilation of 47 y of stream survey data collected within Kansas. Pertinent literature included publications that provided a total number of sites sampled, indicated the number of sites at which Plains Minnow were present, and provided dates and locations of sampling events. If the aforementioned data were included within the publication, proportion of sites occupied by Plains Minnow by year within each basin studied was calculated and added to the dataset. There was unevenness in sampling within the dataset as not all years were sampled, number of sites sampled among basins and years was variable, and some sites were sampled multiple times

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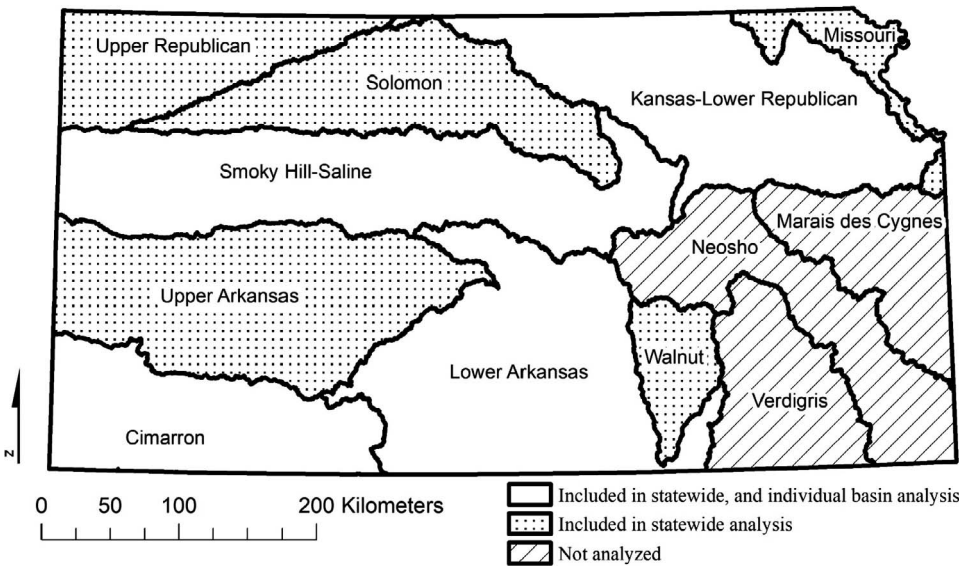


FIG. 1.—Map of Kansas showing the various river basins within the state and the level to which each basin was analyzed. Cimarron river basin (37.322, –100.751); Missouri river basin (39.758, –95.178); Lower Arkansas river basin (37.717, –97.729); Upper Arkansas river basin (38.087, –99.949); Walnut river basin (37.508, –96.941); Kansas-Lower Republican river basin (39.386, –96.103); Smoky Hill-Saline river basin (38.784, –99.678); Solomon river basin (39.566, –99.034); Upper Republican river basin (39.738, –101.483), Verdigris river basin (37.273, –95.724), Neosho river basin (38.025, –95.556), Marais des Cygnes river basin (38.331, –95.022).

between 1964 and 2017, whereas some sites were only visited once. Sites were typically between 150–300 m in length and were sampled by seining, backpack electrofishing, or a combination of the two. Across Kansas sampling occurred during 36 y and across a 47 y period. There was a total of 3837 unique sampling events throughout this time period.

Definitions of river basins were taken from the Kansas Department of Health and Environment (KDHE) surface water register map (Kansas Department of Health and Environment, 2013) (Fig. 1). The Spring River Basin, while within the historical range of Plains Minnow in Kansas, was not analyzed as the species is known from the basin within the state from only one collection and has not been collected in the basin within the last 30 y (Taylor and Eberle, 2014). Data from the Missouri, Solomon, Upper Republican, Upper Arkansas, and Walnut river Basins were not analyzed individually, as either no Plains Minnow were detected within the basins during the time interval studied or there was not sufficient data to run analyses. The Cimarron, Kansas-Lower Republican, Lower Arkansas, and Smoky Hill-Saline River Basins were analyzed individually for trends of proportion of sites occupied over time.

DATA ANALYSIS

We included both historical and new data in our analysis in order to replicate the analysis of historical data which has been conducted by other researchers (Haslouer *et al.*, 2005; Gido *et al.*, 2010), and to allow for better incorporation of any cyclical population variations. Data were summarized as the presence or absence of Plains Minnow at each site, and the proportion of

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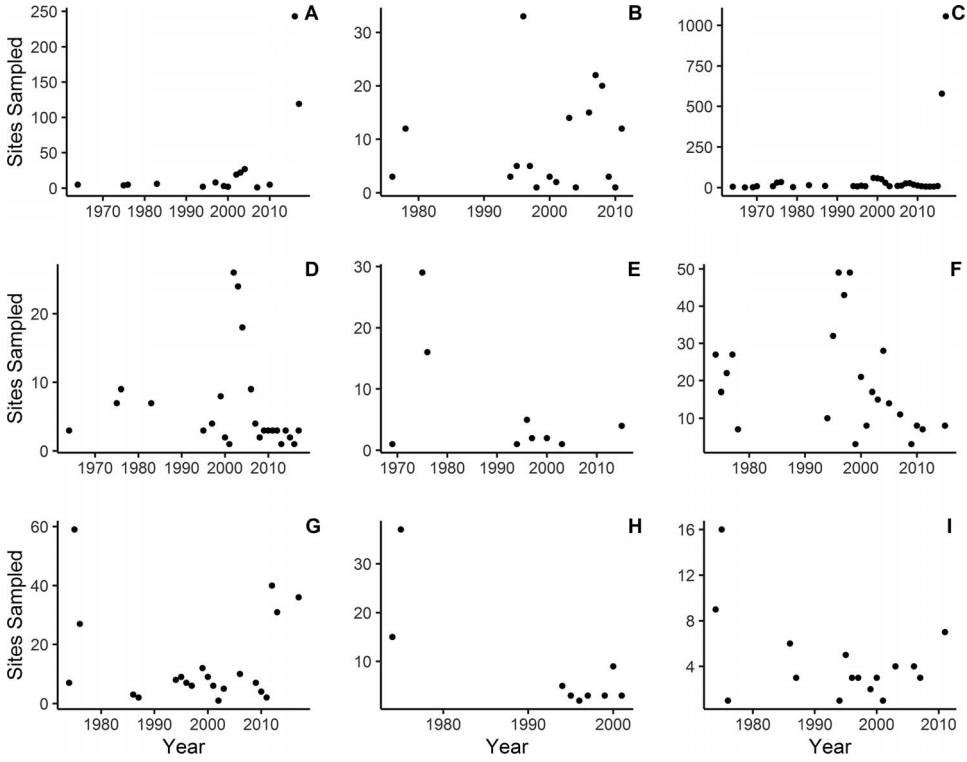


FIG. 2.—Variation of sampling effort within basins. (Panel **A**) Cimarron river basin. (Panel **B**) Missouri river basin. (Panel **C**) Lower Arkansas river basin. (Panel **D**) Upper Arkansas river basin. (Panel **E**) Walnut river basin. (Panel **F**) Kansas-Lower Republican river basin. (Panel **G**) Smoky Hill-Saline river basin. (Panel **H**) Solomon river basin. (Panel **I**) Upper Republican river basin

sites with Plains Minnow present over time was analyzed. To correct for variation in sampling effort (Fig. 2), data were condensed by the multiple sites across multiple streams within basins. All analyses and graphs used program R (version 3.4.3, R Core Team, 2017) and libraries ggplot2 (Wickham, 2016) and gam (Hastie, 2013) with a binomial link = "logit". The critical value used to assess statistical significance was $P < 0.05$. We ran both parametric linear regression models and nonparametric localized regression (LOESS) models. For LOESS models the alpha value (α) was estimated using the gam function, and values above and below the selected α value from the gam model were assessed using the loess function. This evaluation was conducted to ensure the α value would be large enough for sufficient degrees of freedom to represent the underlying pattern, but not overly large to result in a loss of information. We also chose α values to maintain computational efficiency without running into the issue of insufficient data (Takezawa, 2006; R Core Team, 2012; Wood, 2017). The α value gives the percentage of total observations used in each local regression performed. Therefore, α is specified as a value between 0 and 1, and an α value of 0.50 would indicate each local regression incorporates 50% of the total data points. We tested linearity using Analysis of Variance (ANOVA), comparing generalized additive models (GAMs) using a linear or LOESS model. This ANOVA test is a likelihood ratio test between the two models.

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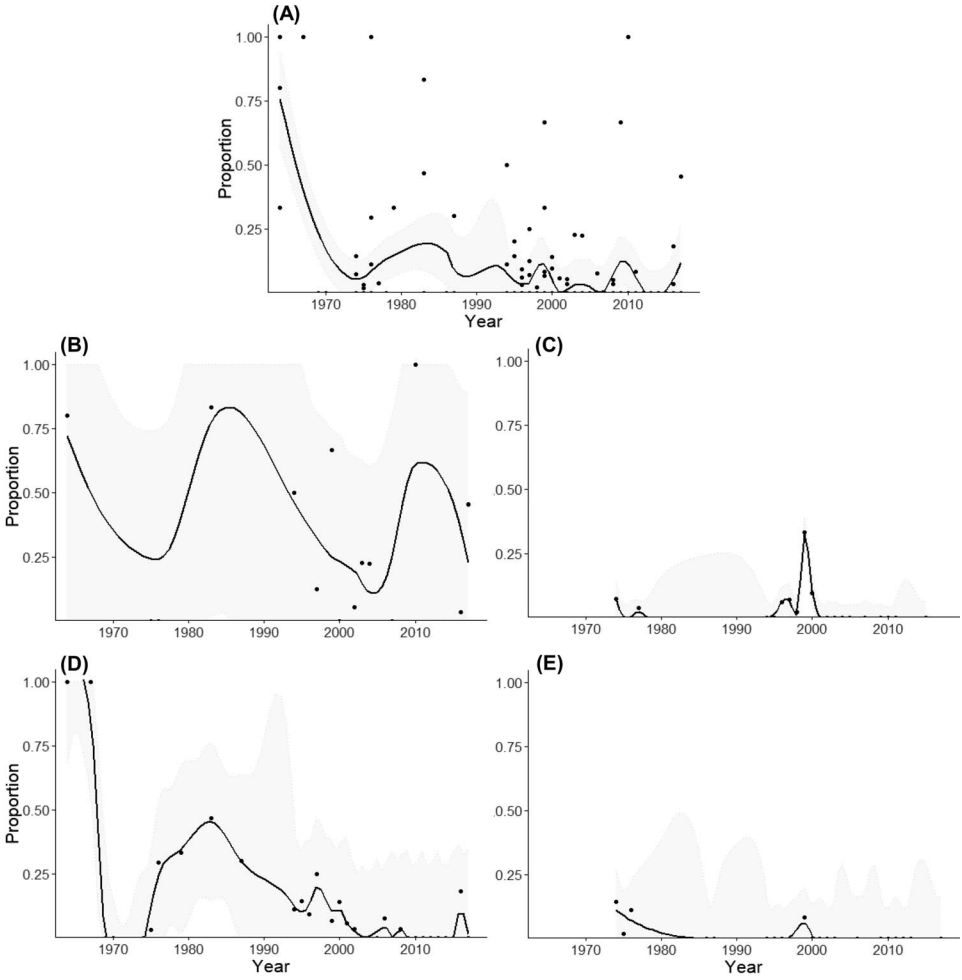


FIG. 3.—Proportion of sites occupied by plains minnow across river basins in Kansas. Shaded areas are 95% confidence intervals. (Panel **A**) Data from the entire state of Kansas and uses nonlinear local regression ($\alpha = 0.1413$). (Panel **B**) Data from Cimarron river basin and uses nonlinear local regression ($\alpha = 0.5419$). (Panel **C**) Data from Kansas-Lower Republican river basin and uses non-linear local regression ($\alpha = 0.0358$). (Panel **D**) Data from Lower Arkansas river basin and uses nonlinear local regression ($\alpha = 0.1618$). (Panel **E**) Data from Smoky Hill-Saline river basin and uses non-linear local regression ($\alpha = 0.0199$).

RESULTS

ACROSS THE STATE OF KANSAS

Sites across the state of Kansas were analyzed by year from 1964 to 2017 (N = 35; Fig. 3A). Results from the ANOVA comparison of the linear versus LOESS models indicated there was a difference in model performance, wherein the LOESS model performed better ($\alpha = 0.1413$; $P < 0.01$; Table 2). The LOESS model indicated parametric and nonparametric relationships for proportion of occupied sites over time ($F_{1,32} = 22.57$, $P << 0.001$; nonparametric $F_{1,32} =$

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TABLE 2.—ANOVA comparing performance of linear versus non-linear models. All models use year as the independent (x) variable and proportion of occupied sites as the dependent (y) variable. Nonlinear models assume year is not linear. Degrees of freedom (df), residual deviance (rd), and p-values (p) are reported. Significant differences ($P \leq 0.05$) in model performance are indicated with an asterisk (*) and the best model is indicated with the double dagger (‡)

Basin	Model	df	rd	P
State of Kansas	linear	35	7.763	
	nonlinear‡	32	5.472	0.001*
Cimarron	linear	13	9.322	
	nonlinear	10	6.873	0.2105
Kansas-Lower Republican	linear	20	2.215	
	nonlinear‡	17	0.706	<<0.001*
Lower Arkansas	linear	30	8.463	
	nonlinear‡	27	5.276	0.0002*
Smoky Hill-Saline	linear	19	0.644	
	nonlinear‡	16	0.338	0.0015*

5.05, $P = 0.006$). A negative relationship was found and indicated that the proportion of sites occupied by Plains Minnow within the state of Kansas has decreased over time. Across the state of Kansas, the proportion of sites occupied ranged from 0 to 1, and the mode was 0. The proportion of sites occupied was highest before 1970, and exhibited a slight increase in the early 1980s, after which proportions were at or near 0 (Fig. 3A).

CIMARRON

The Cimarron River basin was analyzed by year from 1964 to 2017 ($N = 15$; Fig. 3B). Results from the ANOVA comparison of the linear versus LOESS models indicated there was no difference in model performance. The LOESS model did not find a pattern for proportion of occupied sites over time ($F_{1,10} = 0.2261$, $P = 0.64$; nonparametric $F_{1,10} = 1.4749$, $P = 0.28$). Results indicate that the proportion of sites occupied by Plains Minnow within the Cimarron River basin has been consistently fluctuating stable over the time period studied. The highest proportion of sites occupied within a year was in 2010. The proportion of sites occupied fluctuated during the timeframe studied with peaks occurring around 1985 and 2010 while lower proportion of sites occupied occurred around 1975 and 2000 (Fig. 3B).

KANSAS-LOWER REPUBLICAN

The Kansas-Lower Republican River basin was analyzed by year from 1974 to 2015 ($N = 20$; Fig. 3C). Results from the ANOVA comparison of the linear versus LOESS models indicated there was a difference in model performance, where the LOESS model performed better ($\alpha = 0.0358$; $P < 0.01$; Table 2). The LOESS model indicated no parametric relationship, but did indicate a negative nonlinear relationship for proportion of occupied sites over time ($F_{1,17} = 2.8972$, $P = 0.12$; non-parametric $F_{1,17} = 7.307$, $P = 0.002$). The results of these analyses indicate that the proportion of sites occupied by Plains Minnow over time is stable within the Kansas-Lower Republican River basin. However, the species is found infrequently within the basin as the proportion of sites occupied over time ranged from 0 to 0.33 with a mode of 0 and has been at or near 0 throughout the majority of the time period (Fig. 3C). Plains Minnow were last collected within the basin in 2000.

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LOWER ARKANSAS

The Lower Arkansas River basin was analyzed by year from 1964 to 2017 (N = 20; Fig. 3D). Results from the ANOVA comparison of the two models indicated there was a difference in model performance, where the LOESS model performed better ($\alpha = 0.1618$; $P < 0.01$; Table 2). The LOESS model indicated negative parametric and nonparametric relationships for proportion of occupied sites over time ($F_{1,27} = 21.08$, $P < 0.001$; nonparametric $F_{1,27} = 6.1872$, $P = 0.002$). The negative relationships determined by this model indicate that the proportion of sites occupied by Plains Minnow within the Lower Arkansas River basin has declined over time. Proportion of sites occupied was high (1.0 in 1964 and 1965) at the beginning of the timeframe studied but crashed around 1970, rebounded slightly during the 1980's, and has gradually declined since (Fig. 3D). Plains Minnow have been found within only the Salt Fork Arkansas and Medicine Lodge rivers since 2014 despite intensive sampling by the KDWPT Stream Survey.

SMOKY HILL-SALINE

The Smoky Hill-Saline River basin was analyzed by year from 1974 to 2017 (N = 20; Fig. 3E). Results from the ANOVA comparison of the two models indicated there was a difference in model performance, where the LOESS model performed better ($\alpha = 0.0199$; $P = 0.002$; Table 2). The LOESS model indicated negative parametric and nonparametric relationships for proportion of occupied sites over time ($F_{1,16} = 14.69$, $P = 0.001$; nonparametric $F_{1,16} = 3.6077$, $P = 0.037$). The negative relationships determined by these models indicate that the proportion of sites within the Smoky Hill-Saline River basin in which Plains Minnow are present have declined over time. Within the basin, the highest proportion of sites occupied within a year was 0.14 in 1974 and ranged from 0 to 0.14 with a mode of 0. Similar to the trend found within the Kansas-Lower Republican river basin, the proportion of sites occupied over time has been at or near 0 throughout the duration of the timeframe within this basin. The species has been absent from collections within the basin since 1999 and has only been collected once within the basin since 1976 (Fig. 3E).

DISCUSSION

CURRENT STATUS OF PLAINS MINNOW IN KANSAS

It is important to note the models generated in our study are only estimations with high levels of uncertainty, as illustrated by the confidence intervals (shaded area) within Figure 3. Given the uncertainty of the models, we can only draw very basic generalizations from the results. Nonetheless, the findings of our study are consistent with multiple others (Cross *et al.*, 1985; Haslouer *et al.,* 2005; Gido *et al.,* 2010; Perkin and Gido, 2011; Perkin *et al.*, 2015b) and add to the growing body of evidence that the Plains Minnow has been declining within Kansas over the last 50 y (Table 1). Our study supports the findings of Eberle *et al.* (1997), which indicated that Plains Minnow experienced a decline within the Smoky Hill-Saline basin, and supports the conclusions of Perkin *et al.* (2015b), who predicted continual decline and possible extirpation of the species within the Lower Arkansas basin. The Plains Minnow has not been collected within the Arkansas River mainstem since 2015, and it is likely that the species exists only within the Medicine Lodge and Salt Fork Arkansas rivers within the basin (R. Waters, pers. comm.).

The Plains Minnow, once common throughout the state, is now found regularly only within the Cimarron, Salt Fork Arkansas, and Medicine Lodge rivers in south-central Kansas (Gido *et al.,* 2010; Taylor and Eberle, 2014). Intensive sampling by the KDWPT Stream

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Survey in 2016, 2017, and 2018 did not detect Plains Minnow within the Smoky Hill-Saline or Kansas-Lower Republican river Basin, and were absent from the Lower Arkansas River basin, except for within the Medicine Lodge and Salt Fork Arkansas rivers. Additionally, Plains Minnow were not collected during extensive sampling of the Arkansas and Ninnescah rivers in South Central Kansas by Perkin *et al.* (2015b). It is likely that the species is extremely vulnerable to extirpation within the Kansas-Lower Republican basin, as the proportion of sites occupied by Plains Minnow has been near zero during the time period of 1974–2015. Within the near future, there is a real possibility that Plains Minnow will be extirpated from all waterways in Kansas, except for the Cimarron, Salt Fork Arkansas, and Medicine Lodge rivers of south-central Kansas. While these waterways are currently a stronghold for the species within the state, a period of extreme drought, or fragmentation of these systems, could potentially eliminate Plains Minnow from the region.

CAUSES OF DECLINE

Within Kansas it is probable that a combination of three stressors have led to the decline of Plains Minnow within the state as a whole. First, and likely most detrimental, is fragmentation due to damming (either for flood control, water retention, or beautification) and dewatering. Fragmentation results in the gradual disappearance of pelagic broadcast-spawning cyprinids from the river reach that is affected (Perkin and Gido, 2011). Dewatering results in the elimination of all fish species from the affected reach. In a study by Perkin *et al.* (2015b), the mechanism that is seemingly causing the decline of Plains Minnow in these fragmented and dewatered river reaches is described as an "Ecological Ratchet". Regarding the loss of native species, the proposed ratchet works in the following steps: (1) desiccation of a stream reach eliminates a species from the reach, and (2) all attempts of upstream repatriation are blocked by dams, which stop upstream fish movement into the previously desiccated stream reach (Perkin *et al.*, 2015b). It is reasonable to assume that the extirpation of Plains Minnow from the Arkansas River upstream of Wichita, Kansas followed a similar pattern of desiccation, decline, and eventually extirpation due to the dewatering of the river in western Kansas and the dams present on the Arkansas River in the cities of Wichita and Great Bend (R. Waters, pers. comm.).

Second, the disruption of the natural flow regime (Poff *et al.*, 1997) by flood water retention in large man-made reservoirs eliminates the variable flow in which the Plains Minnow evolved to survive and reproduce (Sliger, 1967; Lehtinen and Layzer, 1988; Durham and Wilde, 2006). Multiple studies have illustrated that high flow events synchronize Plains Minnow spawning (Cross and Moss, 1985; Eberle, 2007; Patton and Hubert, 1993). The absence of these events may decrease the reproductive output of Plains Minnow populations, as conditions for reproduction are not ideal. In addition, flooding is vital for the creation of backwater habitats (Schmidt *et al.*, 2001) in which Plains Minnow have been observed aggregating to spawn (Cross, 1967).

Third, channelization alters flow and habitat within the impacted stream reach and leads to the loss of spawning habitat and the appropriate stream morphology for Plains Minnow to complete their reproductive cycle (Di Tomaso, 1988; Worthington et al., 2014; Pennock, 2017). As Plains Minnow are known to aggregate in areas of lower water velocity to spawn (Cross, 1967; Taylor and Miller, 1990), loss of these habitats is apt to negatively impact the viability of Plains Minnow populations within channelized stream reaches. Channelization is continuously occurring within the region as the result of human interaction with the stream environment (Pennock, 2017), as well as the invasive plant, Saltcedar (*Tamarix spp.*), that also causes the channelization of streams in areas of infestation (Di Tomaso, 1988). Saltcedar has invaded and

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become well established within the Cimarron and Lower Arkansas river basins in Kansas that encompass the last stronghold of Plains Minnow populations within the state.

ATTEMPTS AT MITIGATION

Current and past projects that may be of benefit to the reestablishment of Plains Minnow in Kansas include the erection of a fish passage at the Lincoln Street Dam in Wichita, Kansas and repatriation efforts by the KDWPT. The fish passage became operational in 2015 and was the subject of a study by Pennock *et al.* (2018). It was determined that the fishway may aid in restoring connectivity between the segments of river upstream and downstream of the dam (Pennock *et al.*, 2018). However, while the fishway may allow for increased connectivity between these segments, fishes that manage to ascend the passage must then travel through 3 rkm of lentic habitat until reaching lotic habitat. Furthermore, Pennock *et al.* (2018) showed that within the impounded river segment directly upstream of the dam, there was a higher abundance of predatory fishes, such as Largemouth Bass (*Micropterus salmoides*), Green Sunfish (*Lepomis cyanellus*), and Flathead Catfish (*Pylodictis olivaris*), than within other river segments. Additionally, although the fish passage may allow for upstream migration above the dam, any attempts at further upstream migration are halted 9 rkm above the fish passage by another dam, which is not equipped with a fish passage.

Given the fact that Plains Minnow have been extirpated from large portions of the species' native range within the state of Kansas, repatriation efforts may be necessary in order to re-establish viable populations within these areas (Perkin *et al.*, 2015b). In recent years, the KDWPT has attempted to repatriate Plains Minnow within the South Fork Ninnescah and mainstem Arkansas rivers by relocating hundreds of Plains Minnows from the Salt Fork Arkansas River basin into these historically inhabited river segments. These attempts at repatriation have not yet been reevaluated to determine if they were successful or not. Although these attempts at repatriation may create populations of Plains Minnow in these river reaches, they do not necessarily represent a long-term solution. There have been no large-scale attempts to mitigate the impacts of fragmentation, dewatering, and channelization within these river reaches. The conditions that caused the original decline of the species in the area are still present and likely to intensify as climatic conditions are expected to become warmer and drier within the southern Great Plains (Covich *et al.*, 1997). If the conditions causing the decline of Plains Minnow in these areas are still present at the time of repatriation, it is likely the repatriated populations will eventually decline and disappear, no matter the original apparent success, similar to the decline in the original populations.

FUTURE CONSERVATION EFFORTS AND REVERSING THE DECLINE

With the opening of the Kansas Aquatic Biodiversity Center (KABC) at the Farlington Fish Hatchery in southeast Kansas in 2018, the state is uniquely positioned to launch a large-scale repatriation effort for the species. However, for repatriation attempts to be successful in the long term, efforts to mitigate the factors that originally led to the decline of the species in the region need to be made before repatriation. Attempts at mitigation and habitat restoration would need to focus on the restoration of the connectivity of waterways within the region, improved water use practices to limit dewatering, and elimination of channelization (Perkin *et al.*, 2015b; Worthington *et al.*, 2018). If restoration of these habitat factors is attempted, other imperiled pelagic-spawning cyprinids (such as the Peppered Chub and Arkansas River Shiner, both of which are extirpated in Kansas; Worthington *et al.*, 2018) may benefit as well as the Plains Minnow. Restoration of the rivers

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of Kansas would also be beneficial to a wide variety of other aquatic organisms, including freshwater mussels, which are considered by some to be the most highly imperiled group of freshwater organisms (Williams *et al.*, 2003; Propst and Gido, 2004; Archdeacon and Remshardt, 2012; Hitt *et al.*, 2012; Perkin *et al.*, 2015b)

If Plains Minnow are to continue to persist within Kansas, action must be taken in order to reverse the degradation of habitat that has caused the decline of the species. Other species, such as the Peppered Chub, have experienced declines mirroring those that Plains Minnow are currently facing and are now extirpated within the state of Kansas (Perkin *et al.*, 2015b; Pennock *et al.*, 2017; Worthington *et al.*, 2018). Diversity is essential to the resilience and persistence of our global fish communities. Diverse ecosystems are able to recover faster and be more resilient in times of disturbance and environmental change (Olden *et al.*, 2004). With the pressing problem of global climate change, and the multitude of stressors impacting aquatic ecosystems, it is necessary that we make every attempt possible to conserve native fish diversity in order to avoid the biotic homogenization of our waters and the loss of unique species within the environment.

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